



Emergency Alerting Wheelchair for Disabled People with Multiple Control Options

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INTRODUCTION

Our project aims to create a prototype wheelchair that enhances user independence and safety through versatile movement controls, including joystick, voice commands, and traditional methods. It incorporates an integrated help/emergency alerting system, allowing users to request assistance from designated contacts and automatically alerting healthcare facilities in case of accidents, providing real-time patient location.

OBJECTIVE

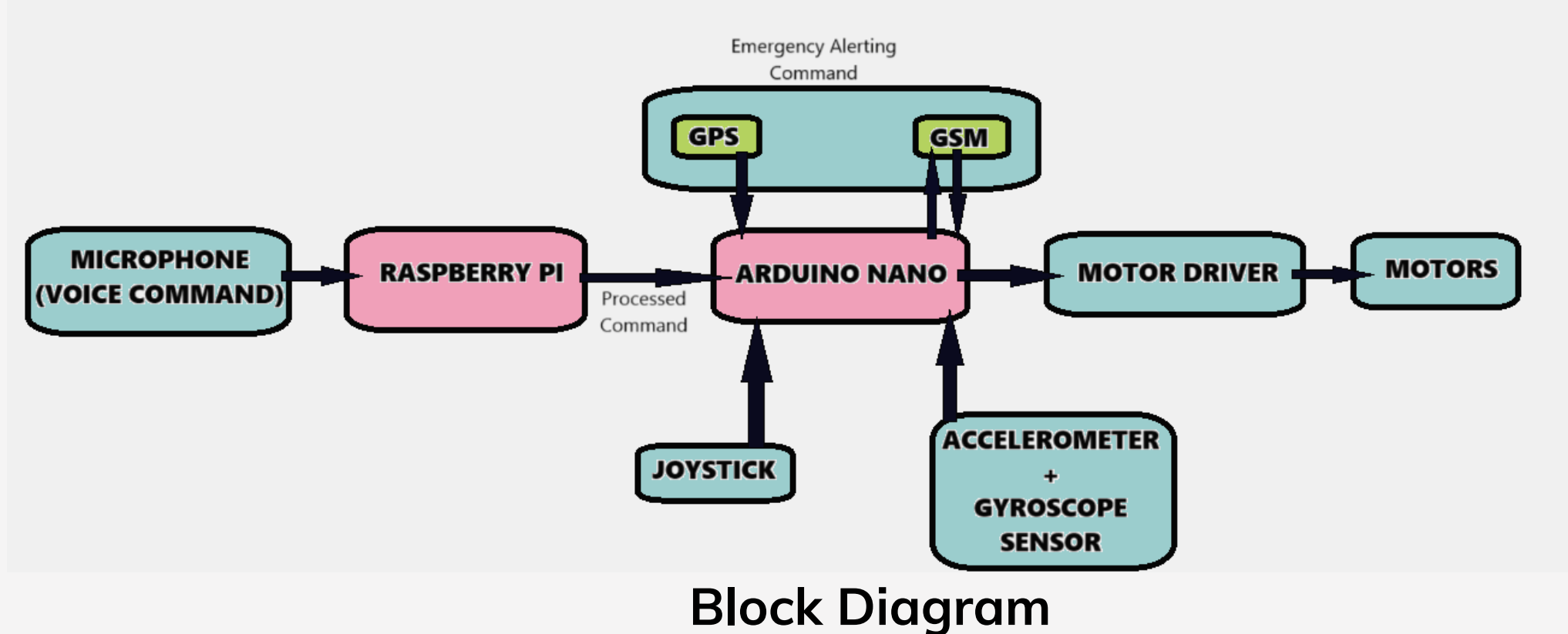
- Wheelchair Movement Control using Voice or Joystick Commands.
- Request Message Feature.
- Emergency Alert System.
- Location Tracking



NEED OF THIS PROJECT

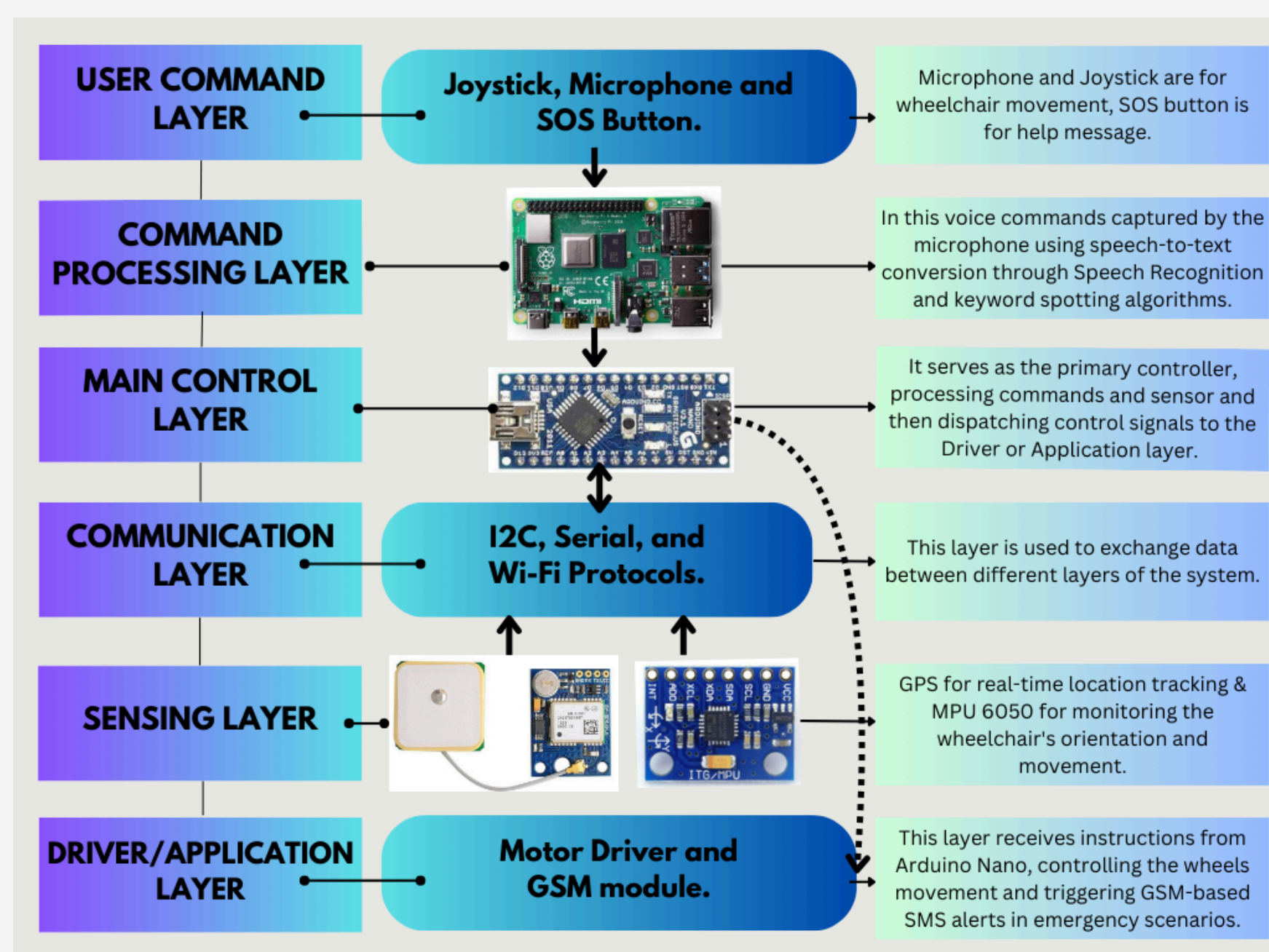
The need of accessible and safe mobility solutions which provide independence for individuals with disabilities and medical challenges in our community. This project seeks to provide a tailored solution that not only enhances their mobility but also ensures their safety and independence, offering vital support in situations unique to their circumstances.

METHODOLOGY

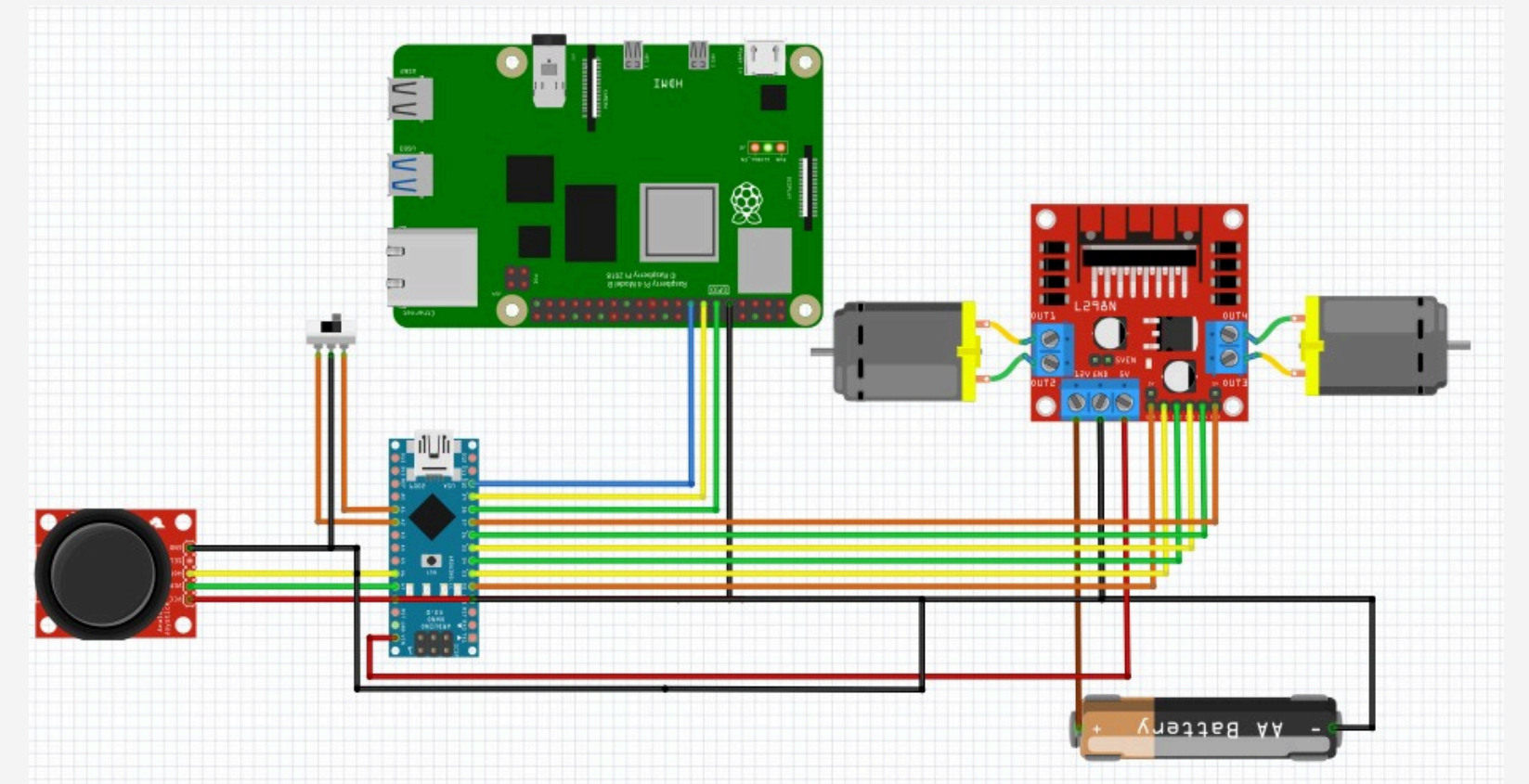


Block Diagram

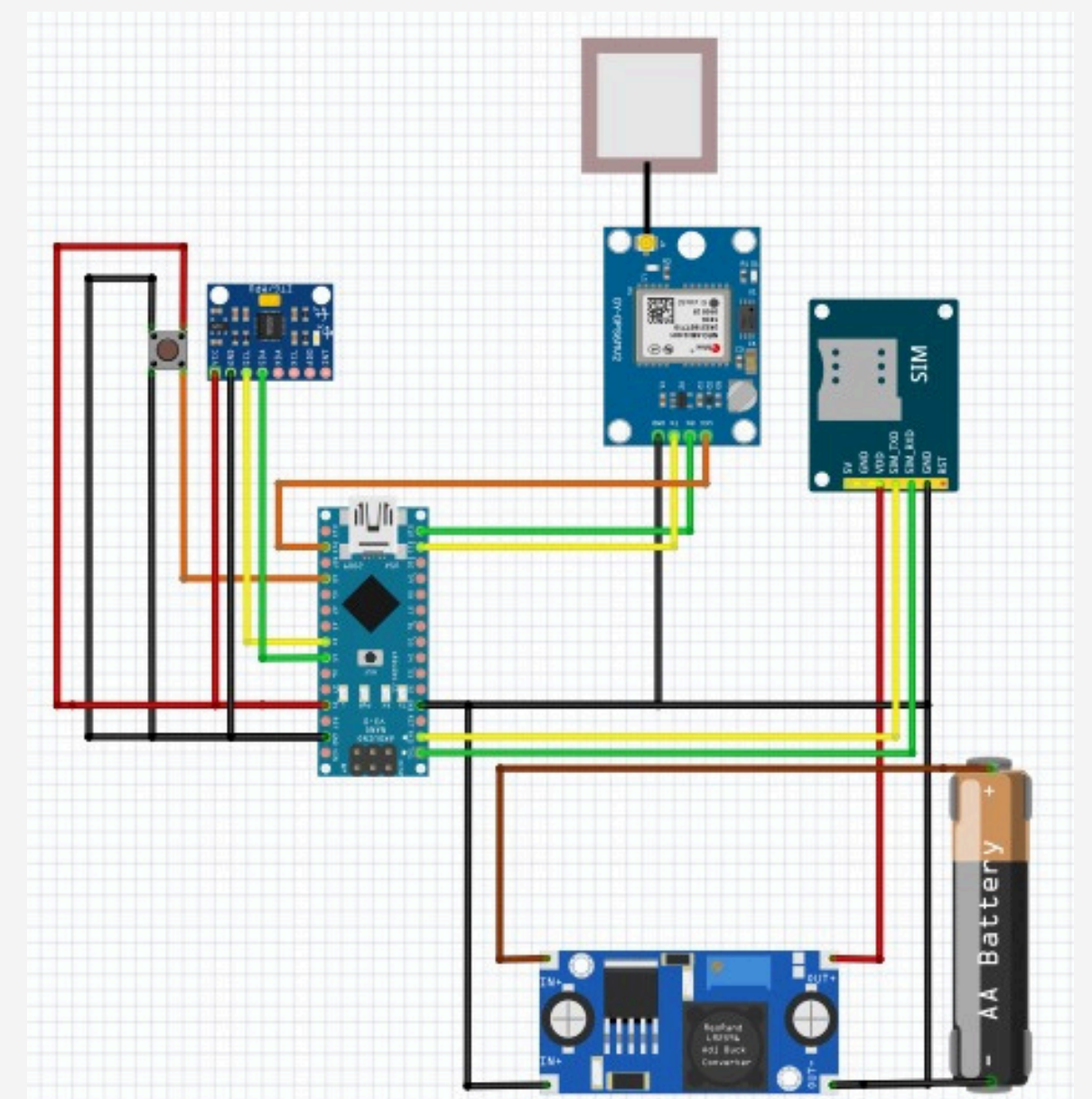
The system offers dual control methods - joystick and voice inputs - alongside emergency alerting features such as an SOS button and GPS tracking. The system architecture comprises five layers, namely User Command, Command Processing, Main Control, Communication, and Sensing, and the driver or application layer. Below shown the layered diagram of the project.



CIRCUIT DIAGRAMS

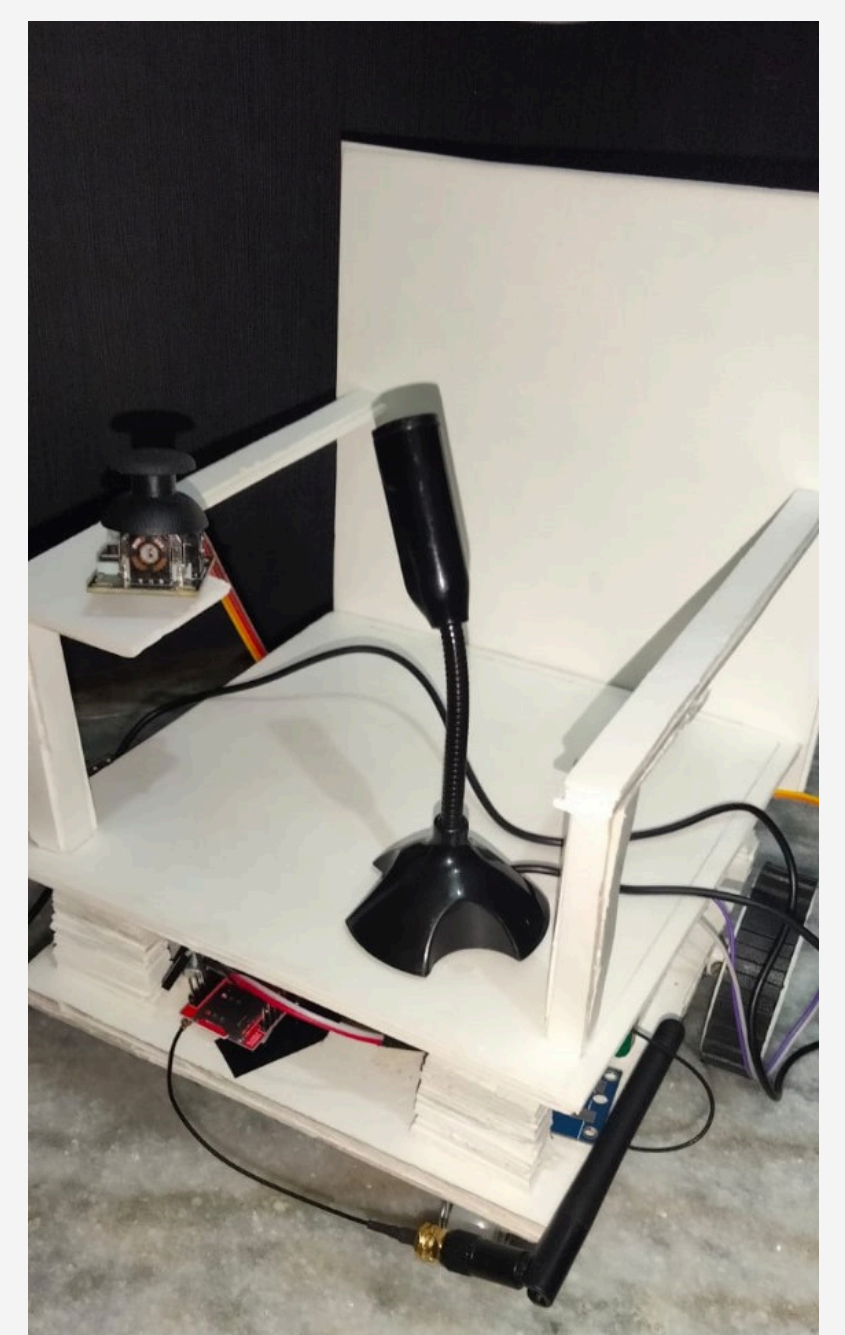
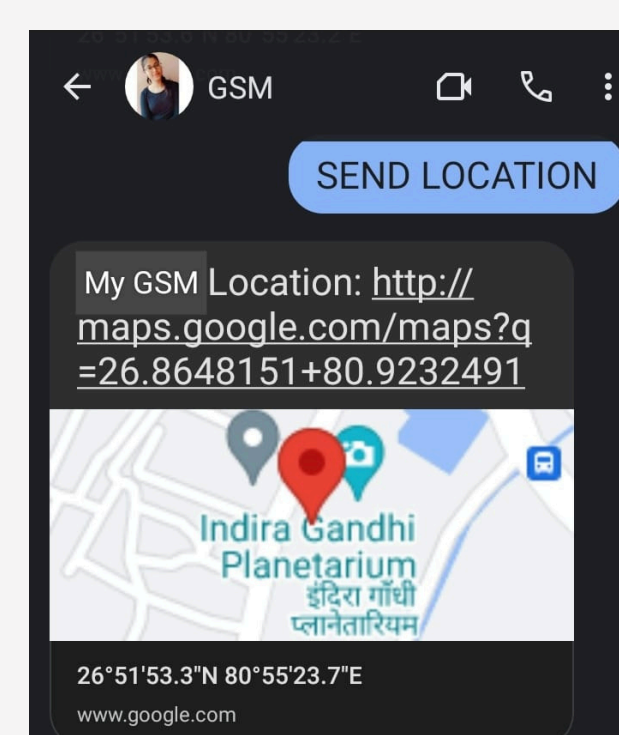
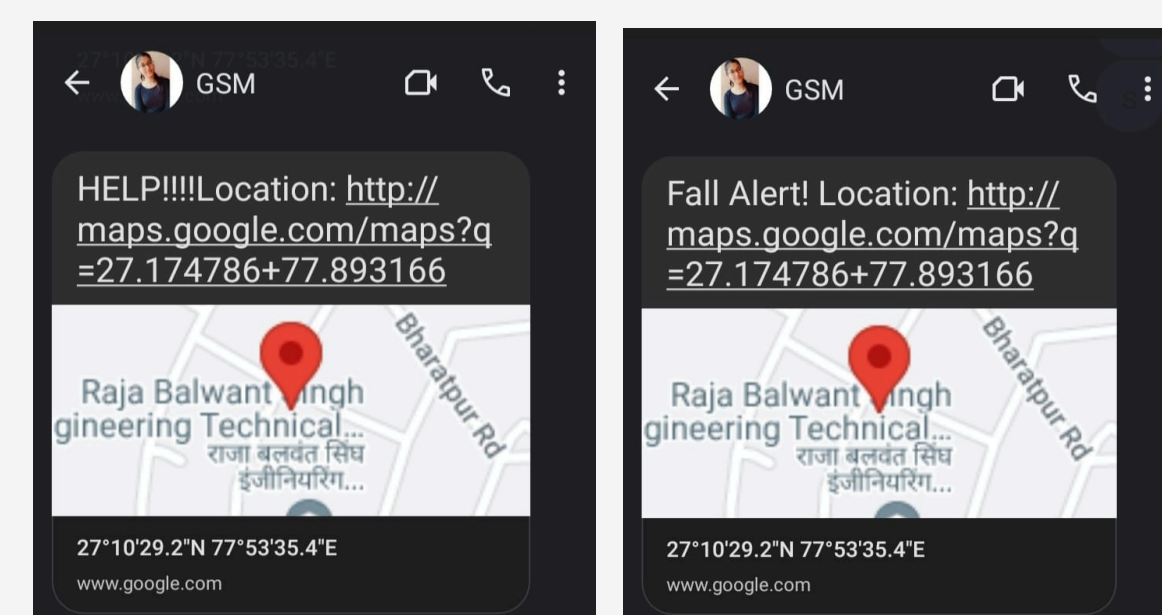


Circuit Implementation of Movement Control



Circuit Implementation of Emergency Control and Location Tracking

RESULTS



FUTURE SCOPE

- Offline Voice Recognition Implementation.
- Enhanced Safety Features.
- Speed Control Features.

APPLICATIONS

- Independent Mobility.
- Outdoor Mobility.
- Elderly Care.
- Emergency Situations.